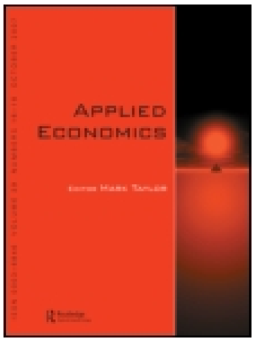

Applied Economics

ISSN: 0003-6846 (Print) 1466-4283 (Online) Journal homepage: www.tandfonline.com/journals/raec20

A blueprint for success in the US film industry

Stephanie M. Brewer, Jason M. Kelley & James J. Jozefowicz

Applied Economics, 2009, **41**, 589–606

A blueprint for success in the US film industry

Stephanie M. Brewer, Jason M. Kelley and James J. Jozefowicz*

Department of Economics, Indiana University of Pennsylvania, 213 McElhaney Hall, Indiana, PA 15705

This article analyses motion picture box-office gross revenue using a cross-section of films from 1997 to 2001. The dependent variable is total domestic box-office revenue. The independent variables investigated include: production budget; peak number of screens that the film was shown on in theaters; Consumer price index for movie tickets; personal income; season and year of the release in theaters; a measure of pre-existing audience; aggregate critic rating; *MPAA* rating; genre; word-of-mouth recommendation; the presence of popular stars and the award nominations. A distinction is made in the analysis between information available to the public prior to the release of the film in theaters (*ex ante*) and information available to the public after the film opens in theaters (*ex post*). Results for the *ex ante* ordinary least squares (OLS) regression reveal positive impacts of budget, summer and holiday release dates, critical reviews, sequels and several genres on gross revenue. Significant, positive determinants in the *ex post* OLS regressions include budget, the peak number of screens, sequels, critical reviews, summer and holiday releases, word-of-mouth, award nominations and star power.

I. Introduction

A differing assortment of films is released to the American viewing public every Friday. While some movies generate immediate box-office success and remain in the theaters for many weeks, others are quickly removed from the screens and promptly replaced with new box-office hopefuls. Cinema statistics provide little comfort to film producers: 'of any 10 major theatrical films produced, on average 6 or 7 are unprofitable, and 1 will break even' (Vogel, 1990, p. 29). With such dim prospects, motion picture production studios are constantly seeking to discover the formula for financially successful films.

Fortunately for film producers, an eager group of scholars is devoted to studying the motion picture phenomenon. From the perspective of the economist, films possess inherent unique qualities that present a compelling subject for study. Smith and Smith (1986, p. 501) paved the way for future study when they first describe motion pictures as 'one of the best examples of the differentiated product envisioned in conventional models of monopolistic competition'. De Vany and Walls (1997, p. 783) described the motion picture industry as 'fascinating . . . whose unusual and complex features challenge economic theory in interesting ways'. Despite the assertion that 'nobody knows what makes a hit or when it will happen', (De Vany and Walls, 1996, p. 1493) a growing number of economists have accepted the challenge.

Studies have begun to clarify one of the largest issues of film revenue prediction: demand for a film

*Corresponding author. E-mail: James.Jozefowicz@iup.edu

Applied Economics ISSN 0003–6846 print/ISSN 1466–4283 online © 2009 Taylor & Francis
http://www.informaworld.com
DOI: 10.1080/00036840601007351

cannot be entirely predicted prior to its release. Films have been described as infinitely variant (Walls, 2004) and this presents an interesting problem for producers. Studios must not identify only the determinants of financially successful films; they must also identify the characteristics that can be controlled for prior to release, and which qualities are not revealed until it is too late. The goal of this article is to construct a model that will analyse the financial performance of films, as measured by gross-domestic revenue, based on a number of readily available indicators. The indicators have been divided into *ex ante* and *ex post* models, in an effort to demonstrate how well success can be predicted before and after the lights have dimmed on opening night.

Some of the indicators present in this study have been largely overlooked by previous articles. In particular, we extend the literature by including an explicit measure of the word-of-mouth transfer of information between consumers. This study has also vastly improved the construction of economic variables intended to represent the demographics of the existing audience and employs a new method of stardom measurement. The model is also comprised of variables that measure the effect of sequels, movie critic reviews, and award nominations, while controlling for the release season, year, genre, *MPAA* rating, production budget, and number of screens. The econometric analysis proceeds via ordinary least squares (OLS) regressions, which are free of classical assumptions violations, such as heteroskedasticity and nonnormality of the residuals, as common in previous research (Prag and Casavant, 1994; Bagella and Becchetti, 1999).

II. Literature Review

Smith and Smith (1986) is the first study to examine the performance of films. They analysed the determinants of successful films, as defined by all-time box-office revenues. Their regressions support the Spence-Owen-Waterman hypothesis that films have become increasingly more specialized as a result of television. Prag and Casavant (1994) found a positive impact of stardom, ‘quality’, sequels and Academy Awards on revenues when marketing expenditures are absent. Stardom, Academy Awards, and production costs are found to be positive determinants of advertising spending. Sawhney and Eliashberg (1996) researched the adoption of motion pictures within the industry. They define the adoption process in terms of both the time to decide to see a movie and the time to act on the given decision. The models only contain variables that were available before the film’s release. It is determined that is it is easier to predict the box-office success of films with no box-office data than to predict the time periods, in which the revenues will be collected.

De Vany and Walls (1997, p. 784) employed a large sample of motion picture revenues and theater bookings throughout theatrical runs in an attempt to model the competition among films as ‘an evolving rank tournament of survival’. They discover that the failure rate of movies is dependent upon time and survival time is related to the number of initial bookings. Long-runs do not guarantee success because revenue is found to be convex. Blockbuster films have unusually extended and wide play lengths, while the majority of films earn modest revenues early before quickly fading away.

Fernández and Baños (1997, p. 61) studied cinema demand in Spain as measured by the average cinema attendance per inhabitant per year. It is stated that the demand side of the film exhibition industry is more powerful than supply because the consumer dictates the market. Theater tickets are sold at a fixed price and are only constrained by the seat capacity of the theaters. Supply is perfectly elastic at the fixed price until the maximum capacity is realized. It can be implied that the quantity sold in a given period is equal to the quantity demanded because most theaters are not selling out every showing. Via cointegration analysis, movies are found to be luxury goods with elastic price demand.

Bagella and Becchetti (1999) analysed the box-office performance of Italian films with special focus on the *ex ante* popularity of actors and directors. While the net impact of government subsidies on total admissions is revealed to be statistically insignificant, the *ex ante* popularity of actors and directors were both found to exert a significant positive impact on the total number of admissions to native-produced films. An additional measure of the positive externalities generated by the performances of the popular figures is also found to be positive and statistically significant.

Ravid (1999) examined film revenue and return-on-investment (ROI) as functions of production cost and star actors, among other variables. Regressions show that large production costs significantly increase film revenue, but do not increase the ROI. The quantity of critic reviews, which is representative of exposure and availability to the public, is positively significant. Sequels are found to perform significantly better than nonsequels. While univariate tests suggest that movie stars increase revenue, regressions find star presence to be insignificant. This supports the 'rent-capture' hypothesis that stars earn salaries equivalent to their market value and do not impact the profitability of films.

Simonoff and Sparrow (2000) studied the success of films, as defined by the logged-domestic box-office gross revenue. The explanatory variables are divided into *ex ante* and *ex post* models in an attempt to pinpoint exactly how much success can be predicted prior a film's release. A dummy variable is included to control whether the film was made in the United States, in another English-speaking country, or in a nonEnglish-speaking country. The *ex ante* model suggests that skilled actors have a positive and statistically significant influence prior to a film's release. The *ex ante* model suggests that the production budget is insignificant in relation to the total gross revenue.

MacMillan and Smith (2001) studied cinema admissions and supply in Great Britain using annual data spanning five decades. The impact of television ownership on theater attendance is explored because it serves as a close or superior substitute for moviegoers. New specialized stations are able to satiate the preferences of diverse groups of people. Television also exhibits an increase in comfort, convenience and privacy that the theater environment restricts. The findings indicate that television ownership is indeed an ongoing negative demand shock and is partially to blame for the decline in theater admissions.

Kelwick (2002), webmaster of The Usher – Just One Opinion, details five factors of film success: an established audience, presence of major stars, advertising, publicity or word-of-mouth and awards. He argues that an established audience[1] gives a movie a head start on opening weekend and high profile stars have the ability to 'build up the anticipation for a movie (alone)'. Kelwick points out that The Blair Witch Project's internet advertising helped it to become 'one of the most profitable movies of all time'. Word-of-mouth sunk Batman and Robin because of an internet-based 'hate campaign'. Kelwick also argues that winning awards brings additional publicity to otherwise unnoticed films.

Ravid (2004) emphasized the risk associated with making movies by estimating a ROI model. The study focuses on the strategies utilized by studio executives when choosing the films to be released. Unlike many other studies, the effect of violence in R-rated films is examined. The results illustrate that high-violence films are expected to be financially 'safer' to produce. On average, the ROI of violent films falls within the middle of the sample's distribution.

De Vany (2004, p. 49) examined the adaptive nature of the film market by targeting the information flow to consumers. Studios often exploit the 'blockbuster strategy' in an attempt to persuade theater owners to book the film and consumers to attend the showings. Positive 'uninformative information' is spread to the targeted audiences without revealing quality indicators. A big budget film or one with high-profile stars often uses this method to increase the opening-weekend screen count and to inflate the opening weekend gross. The transfer of consumer preferences through word-of-mouth usually lessens following the first week. According to the 'blockbuster strategy', the budget, number of screens in the opening weekend, and presence of major stars are all positive and significant in the regression results.

Walls (2004, p. 94) focused on the mobster film genre. This genre is unique because the 'crime-driven action-adventure storyline(s) can transcend the barriers of language and culture' that other genres face. The Pearson correlation coefficients between the production budget and US domestic and foreign box-office revenues, video/DVD sales, rentals and TV revenue, reveal that domestic success is a strong indication of foreign success. Production budgets are highly correlated (0.70–0.80) with all revenues.

Deuchert *et al.* (2005) focused their study on the impact of Academy Award nominations and wins on the weekly box-office revenues of movies. 'Best picture', 'best actor/actress in a leading role' and 'best actor/actress in a supporting role' are examined. The Oscar nomination dummy variable is set equal to one the week that the film is nominated and for all remaining weeks. The Oscar win dummy is set equal to one the week that the film wins the award and for all remaining weeks. The study concludes that Oscar nominations significantly increase weekly revenues, while wins are insignificant.

Dewenter and Westermann (2005) used aggregate data to examine the German cinema market. They use time-series data from 1950 to 2002 to run OLS regressions for cinema demand and SUR and 2SLS regressions for supply. They find cinema demand to be both price and income elastic, with respective elasticities of -2.25 and 4.48. Additional movies and other leisure activities are found to be substitutes. The number of seats and German movies released each

[1] Harry Potter and the Sorcerer's Stone (book), Tomb Raider (video game), James Bond (series) and Star Trek (franchise) were cited by Kelwick (2002) as examples of films that opened with the support of established audiences.

year are found to have significant positive impacts on demand. When considering cinema supply, lagged population is found to have a significant positive coefficient while both video-cassette recorders and televisions have significant negative coefficients.

III. Data

This study analyses a pooled cross-section of the top 100 domestic grossing films per year, for the years 1997 through 2001, and yields 466 observations (Nash, 2002). The lack of availability of data for the word-of-mouth variable limited the sample to these years. Smith and Smith (1986), De Vany and Walls (1997), Albert (1998) and Deuchert *et al*. (2005) used samples of similarly high-performing films in their studies. All films that were re-released after an initial release in a previous year are excluded from the sample.[2] IMAX films are also excluded from the data set because IMAX screen constraints have prevented their widespread availability.[3] Other outliers removed from the sample are Titanic, due to unusually high revenue, and The Blair Witch Project, for its unusually low budget. Films with missing data observations are also omitted from the sample. A complete listing of the films included in the sample appears in Appendix 1.

Variables and hypotheses

The dependent variable, total domestic box-office revenue of a film (*GROSS*), is measured in millions of real US dollars. According to Prag and Casavant (1994, p. 219), film gross is '(t)he more widely discussed measure of a film's revenue'. Revenue generated by sales and/or rentals of VHS and/or DVDs, revenue generated by a re-releases, and revenue from foreign box office runs is omitted. For these reasons, we believe our dependent variable is not vulnerable to the limitations of monetary measures (i.e. problems with correcting for inflation) noted by Bagella and Becchetti (1999) and incurred by Prag and Casavant (1994).

Each independent variable used in the regressions is explained in the following sub-sections. Following the work of Sawhney and Eliashberg (1996) and Simonoff and Sparrow (2000), a distinction is made between information available to moviegoers before the release of the film (*ex ante*) and information available to prospective audience members after the film's release (*ex post*). See Table 1 for a summary of the hypotheses regarding the expected signs of the independent variables.

Table 1. Summary of hypotheses

Independent variable	Expected relationship
log(*BUDGET*)	Positive
log(*SCREENS*)	Positive
log(*ORIGGROSS*)	Positive
log(*INCOME*)	Positive
CPITICKET	Negative
CRITIC	Positive
PREVAUD	Positive
SUMMER	Positive
THANKS/XMAS	Positive
*STARPOWR*2	Positive
CINEMASCORE	Positive
FILMNOM	Positive
Genre variables	Ambiguous
MPAA ratings	Ambiguous
Year dummies	Ambiguous

Variables available prior to release (ex ante)

When seeking successful film characteristics, one could argue that certain genres are inherently more successful than others due to the dissimilar tastes of different groups (Bagella and Becchetti, 1999). One genre may be more likely to prosper than another due to the size, location and leisure preferences of the different target audiences. When holding a light to the box-office numbers, however, evidence shows that this suggestion may not hold to be true. The five highest grossing films of 1997 include two dramas, two comedies, and one action-adventure film.[4] With such a range of genres obtaining similar levels of success, it is difficult to judge if certain genres do have more drawing power than others. The genres included in the model are science fiction (*SCIFI*), comedy drama (*COMDRA*), drama (*DRAMA*), comedy (*COMEDY*), action adventure (*ACTADV*), horror (*HORROR*) and animation (*ANIM*). Each enters the model as a dummy variable, and each is expected to have an ambiguous effect on the *GROSS* variable. The drama genre is the omitted condition.

As discussed by Kelwick (2002), a film is more likely to perform well if it has a built-in audience from a pre-existing fan base. 'Whether they are adaptations of books, comics, plays, computer games

[2] The excluded re-released films include: The Exorcist, Star Wars, Grease and The Little Mermaid.
[3] The documentary, Michael Jordan: To the Max is the excluded IMAX film.
[4] The five highest grossing films of 1997, as reported by The Internet Movie Data Base were: Titanic (drama), Men in Black (comedy), Jurassic Park II: The Lost World (action-adventure), Liar, Liar (comedy) and Air Force One (drama).

or TV shows, to have an audience already in place is a big step towards making your movie a hit', Kelwick argues. A dummy variable measures the effects of a built-in audience (*PREVAUD*). A value of one indicates that a film is a sequel or contains material based on a television show, video game or book. Furthermore, the gross of the film preceding the sequel (*ORIGGROSS*), is included to account for the varying levels of success, on which the sequel has to build. Prag and Casavant (1994) asserted that sequels enable studios to signal information regarding the tastes a particular film seeks to satisfy. De Vany (2004, p. 134) asserted that sequels are produced as replicas of favourable films in an attempt to recreate success. According to Kelwick's theory, both of these variables are expected to have a positive effect.

US personal income for the month of a film's release (*INCOME*) is included in the model. A positive relationship is expected because economic theory states that higher incomes result in increased consumption of normal goods. Movie going is assumed to be a normal leisure activity. According to Becker's (1965) allocation of time theory, a substitution in demand away from time-intensive activities, such as watching movies, may be observed as the shadow price of time rises due to increasing income.

A price-measuring variable is included in the model. Data for the average annual movie ticket prices were obtained from the *MPAA* but could not be used in the model because of collinearity with the year dummies. For this reason, the consumer price index measurement for ticket prices for theatrical plays, movies and concerts for the month of a movie's releases is used (*CPITICKET*). This proxies for movie ticket prices and is expected to have a negative coefficient in accordance with the law of demand.

The judgments of film critics may come into play when determining a movie's box office draw. Moviegoers have easy access to critics' opinions through reviews in news articles, magazines, television and websites. It is unclear whether critics are able to stimulate demand for movies or are just capable of predicting demand. Critical reviews (*CRITIC*) are taken from the website, Rotten Tomatoes, because the aggregate rating is presented in the form of a percentage.[5] Our *CRITIC* variable is an improvement upon the variable in Prag and Casavant (1994), Sawhney and Eliashberg (1996), Eliashberg and Shugan (1997) and Simonoff and Sparrow (2000), who all found critic reviews to have a positive effect on revenue.[6]

De Vany and Walls (1997) observe that film distributors seek high demand periods when selecting a date to release their films. Historically, films are more successful when released during the summer or a holiday season when many moviegoers are on an extended vacation from work or school. The *SUMMER* dummy variable captures films released in May, June or July.[7] The thanksgiving holiday break leads to November success, and a similar effect is witnessed during the December Christmas break. The *THANKS/XMAS* dummy is used to capture films released in November or December. Both *SUMMER* and *THANKS/XMAS* are set equal to one if a film is released in the given time period and is otherwise set equal to zero. The dummies *SUMMER* and *THANKS/XMAS* are expected to show a positive relationship with the dependent variable, *GROSS*. Bagella and Becchetti (1999) averred that the *ex ante* popularity of actors influence the film's success. The variable *STARPOWR* is included to capture the *ex ante* box office drawing power of certain stars. Other researchers such as De Vany and Walls (1999) have utilized lists of Hollywood's 'Most Powerful People' published by Premiere Magazine and James Ulmer's list of A/A+ people. These lists are lengthy and include some individuals who, in our estimation, ought to have substantially less drawing power than others on the list. Prag and Casavant (1994, p. 220) test two different measures of stardom – a dummy for films containing at least one established star and a quantity/quality index for the presence of no stars, a rising or falling star, one established star and more than one established star – based on the authors' 'knowledge of films and movie stars'.

We believe that our approach is an improvement upon the described methods. The *STARPOWR* variable in this study is based upon the annual top ten 'favourite movie star' lists published by the Harris Poll and the annual People's Choice Award nominees for the following categories: motion picture actor, motion picture actress, male television performer and female television performer. There are three People's

[5] The reviews were compiled from a number of respected critics nationwide and the Rotten Tomato rating is the percentage of good reviews the film has received. The same critics review all films available on the site, which suggests constant and unbiased reviews.

[6] Prag and Casavant (1994) concede to popular notion and argue that movie critics' tastes are not necessarily an accurate reflection of the preferences of the general public.

[7] August releases are not included in *SUMMER* because fall semester classes typically resume by the end of the month, resulting in declining ticket sales.

Choice Award nominees for each category. A nationwide poll conducted by the Gallup Organization determines both the nominees and winners. Both polls are well known, established, and cited regularly in the popular press and other research studies. Both polls ask responders open-ended questions to state their favourite person rather than providing a predetermined list of stars from among whom to choose. Every poll involved a sample size between 1000 and 2225 respondents, and responses were weighted to more accurately reflect the average demographics of the US population. We have compiled a master list across the top ten Harris Poll lists for the years 1995–2002 and the People's Choice nominations for the broadcast years 1997–2002. While our movie data is from the years 1997–2001, we believe backdating the Harris Poll lists allows for lagged popularity effects. Due to overlap from year-to-year and across the Harris Poll and People's Choice nominations, our list consists of 66 high profile stars and is included in Appendix 2.

While causality between star popularity and blockbuster-hit movies is potentially an issue, we believe that pooling the polls across years has minimized the effects. Additionally, we believe causality is a larger problem when a previously unknown star is involved in a blockbuster hit that causes his/her popularity to explode. The stars in our master list all experienced success before the year, in which their names appeared in the polls. For this reason, our measure of stardom is *ex ante* in the same sense as Bagella and Becchetti (1999).

The *STARPOWR* variable is a count variable representing the number of individuals from the master list who appear in a given film. Rosen's (1981) superstar phenomenon asserts that market size and revenues in the fields of sports, arts and letters generally favour more highly gifted individuals. According to Bagella and Becchetti (1999, p. 242), a relatively lower *ex ante* popularity substitutes poorly for higher popularity in movies so human inputs should have a nonlinear effect on total admissions. As a result of this interpretation, *STARPOWR* enters our models as a quadratic term, *STARPOWR*2.

MPAA ratings included in the model are *G*, *PG*, *PG*13 and R. According to Prag and Casavant (1994), these ratings emit signals regarding film content that moviegoers find informative in their personal decision-making. Each is set as a dummy variable and is expected to have an ambiguous effect on the *GROSS* variable. The R-rating is the omitted condition as suggested by Jansen (2005). Dummy variables for the years of release are included in the models. Bagella and Becchetti (1999) also used this approach. The year dummy variables, *DUM*98, *DUM*99, *DUM*00 and *DUM*01, are expected to have an ambiguous relationship with *GROSS*. The year 1997 is the omitted condition.

The total millions of real US dollars used to produce a film (*BUDGET*) is considered to be vital because of its ability to proxy other hard-to-find variables. Wages paid to actors and directors as well as the costs for developing special effects and/or obtaining exotic film locations are captured by this variable. Bagella and Becchetti (1999, p. 238) described the movie industry as a place where 'movies are positioned according to their artistic content and to the intensity of their special effects'. They point out that the two opposite extremes of the range are the low capital intensity and high artistic merit film (i.e. 'film d'autore') and the low artistic merit and high capital intensity film (i.e. 'special effects' film). The former is relatively labour-intensive while the latter is relatively capital-intensive.

The dichotomy can be represented by examining two films of equal budgets. The films Unbreakable and Vertical Limit each had budgets of $75 million. Vertical Limit takes place on a snowy mountain and contains costly special effects (e.g. avalanche). One could reasonably assume that much of the film's budget was exhausted on visual manipulations, and qualifies as a 'special effects' film. Unbreakable is the opposite, lacking special effects and exotic filming locations. It instead features bankable stars Bruce Willis[8] and Samuel L. Jackson,[9] with director M. Night Shyamalan,[10] qualifying as a labour-intensive 'film d'autore'. The different expenditures within a production budget further complicate the model, because it is difficult to specify what portion of the budget was used for capital and labour.

The *BUDGET* variable has been omitted in many previous studies. Sawhney and Eliashberg (1996) recognized the possibility of an omitted variable bias due to their lack of a budget variable. Simonoff and Sparrow (2000) included a budget variable in their model, but acknowledged the difficulties that rose when it was found to be missing from the majority of the observations in their dataset. This variable is expected to have a significant positive effect on revenue, as attained by Ravid (1999) and Jansen (2005).

[8] Total domestic box office for Bruce Willis during his career is over $2.1 billion.
[9] Total domestic box-office for Samuel L. Jackson during his career is over $2.8 billion.
[10] M. Night Shyamalan previously directed The Sixth Sense, which earned $293 million domestically.

*Variables available after release (*ex post*)*

The variable *SCREENS* is the maximum number of screens on which a film played. As opposed to the opening number of screens, the maximum is used to represent the availability of the film to potential viewers. De Vany and Walls (1997), Simonoff and Sparrow (2000) and MacMillan and Smith (2001) established the relevance of such a variable in the analysis of film performance. A positive relationship is expected between *SCREENS* and the dependent variable.

Kelwick (2002) and De Vany (2004) stressed the importance of word-of-mouth recommendations as a factor in film performance because they have the potential to spawn bandwagon effects. Cameron (1986) echoes this sentiment by observing that the opportunity cost of viewing a disappointing film can be large after considering the time and effort invested. De Vany and Walls (1997) demonstrated that the information dynamics of movies can produce bandwagon effects, and De Vany (2004) noted that poor word-of-mouth reviews of films can be the 'kiss of death' in revenue rankings even for early revenue leaders. De Vany (2004) acknowledged that word-of-mouth is a bit of a misnomer in the modern world of the internet due to the numerous opportunities for the exchange of private evaluations of films. We implement word-of-mouth information transfer in our model with the poll service cinemascore. Each Friday, cinemascore conducts exit polls in 14 or 15 sites that the company routinely surveys, of 400–500 moviegoers as they leave after viewing a film. The viewers are asked to assign a letter grade to the film they just watched. Cinemascore then aggregates the national scores and posts the results on their website. The grades are translated into the dataset as such: $A+=15$, $A=14$, $A-=13\ldots F-=1$. The variable *CINEMASCORE* is expected to positively affect *GROSS*.

Major award nominations are expected to affect revenues, as suggested by Smith and Smith (1986), Prag and Casavant (1994), Simonoff and Sparrow (2000), Kelwick (2002) and Deuchert *et al.* (2005). Similar categories are observed across four prominent awards: the Academy Awards, the Golden Globes, the (Orange) British Academy of Film Awards and the Screen Actors Guild Awards. The award nominations studied for the model are best picture, best actor, best actress, best supporting actor, best supporting actress, best director and best original screenplay. Prag and Casavant (1994), Simonoff and Sparrow (2000) and Deuchert *et al.* (2005) also examined the significance of nominations, but only in regards to the Academy Awards. In order to prevent collinearity issues between dummy variables for each award, a single award count variable (*FILMNOM*) is created to account for each time a film was nominated for one of the aforementioned award categories. The range for this dummy variable is 0–22. It is expected that the *FILMNOM* variable will be positively correlated to *GROSS*.

Descriptive statistics

Table 2 presents descriptive statistics for the sample. In descending order, the five highest grossing films are Star Wars: Phantom Menace (1999), Harry Potter and the Sorcerer's Stone (2001), Lord of the Rings: The Fellowship of the Ring (2001), The Sixth Sense (1999), Shrek (2001). The five lowest grossing films are Dark City (1998), Excess Baggage (1997), The Messenger: The Story of Joan of Arc (1999), Blues Brothers 2000 (1999) and Les Miserables (1998). The highest grossing movie, Star Wars: Phantom Menace, earned revenues of $431.1 million while the lowest grossing film, Les Miserables, earned $13.9 million. The average gross is $64.35 million dollars with a SD of $55.2 million.

Films receive an average of 50% positive reviews from critics, while reviews from moviegoers have a mean value of 11.27 (B/B+). The difference is imperative because it represents the differing opinions of critics and the general public. Nearly 50% of the films in the sample are rated R by the *MPAA*, which is consistent with results reported by Ravid (2004). He stated that risk-averse studio-heads will opt for the profit-safe topics of sex and violence over the potentially lucrative, but risky genre of children's films. Comedies and dramas make up over half of the films in the sample, while science fiction films compose only 3%.

IV. Model

Classical assumption tests

The regressions were tested for heteroskedasticity via the White test. The results of the test show the inability to reject the null hypothesis of homoskedasticity. The test statistic calculated is 66.20201 and the 5% chi-square critical value is 69.832 with 52 degrees of freedom. The sample does not suffer from heteroskedasticity.

Residuals from the OLS regressions of all models pass normality tests in keeping with the classical assumptions of OLS regression. The null hypotheses of normality for the residuals are not rejected at the 0.5% level (Jarque–Bera test statistics are 0.26 for *ex ante* Model 1, 0.41 for *ex post* Model 2 and 0.59 for *ex post* Model 3).

Table 2. Descriptive statistics

Variable	Mean	SD	Max	Min
GROSS	\$64.35 million	\$55.19 million	431 million	\$13.87 million
BUDGET	\$47.88 million	\$31.84 million	\$200 million	\$1.7 million
SCREENS	2415.8	584.3	3715	510
INCOME	\$7.88 trillion	\$6.39 billion	\$8.73 trillion	\$6.75 trillion
ORIGGROSS	\$3.46 million	\$14.7 million	\$85.7 million	\$1.3 million
CPITICKET	108.7	7.31	120	100
CRITIC	50.77	26.14	100	0
PREVAUD	0.35	0.48	1	0
SUMMER	0.24	0.43	1	0
THANKS/XMAS	0.24	0.43	1	0
STARPOWR	0.39	0.60	4	0
CINEMASCORE	11.27	2.05	15	4
FILMNOM	0.99	2.95	22	0
DRAMA	0.29	0.45	1	0
ACTADV	0.16	0.37	1	0
ANIM	0.06	0.24	1	0
COMDRA	0.12	0.32	1	0
COMEDY	0.27	0.45	1	0
HORROR	0.06	0.24	1	0
SCIFI	0.03	0.18	1	0
G	0.04	0.21	1	0
PG	0.13	0.34	1	0
*PG*13	0.38	0.49	1	0
R	0.44	0.50	1	0
*DUM*98	0.20	0.40	1	0
*DUM*99	0.20	0.40	1	0
*DUM*00	0.20	0.40	1	0
*DUM*01	0.20	0.40	1	0

Final models

Using an approach similar to Simonoff and Sparrow (2000), the variables are used to construct three models. Model 1, the *ex ante* model, only includes variables that are available prior to a film's box-office release. Models 2 and 3, the *ex post* models, include variables available after a film's release in addition to all *ex ante* variables. The *ex post* models could not simultaneously contain log(*budget*) and log(*screens*) due to collinearity. Model 2 includes log(*budget*) as an independent variable while Model 3 includes log(*screens*).

The final estimated models use a double-log functional form. According to Fernández and Baños (1997), the double-log specification is very common when estimating demand functions for unrelated goods and has yielded good results in studies by Moore (1966), Withers (1980) and Abbé–Decarroux (1994, p. 64). The *GROSS* variable is logged due to its 'long right-tailed nature' (Simonoff and Sparrow, 2000). The same logic is applied to the *BUDGET*, *ORIGGROSS* and *INCOME* variables and they enter the equations as logarithms.

The inclusion of the genre and *MPAA* rating variables was confirmed by *F*-tests at the 1 and 5% levels, respectively (*F*-test statistics for genres are 2.94 for *ex ante* Model 1, 5.12 for *ex post* Model 2 and 2.10 for *ex post* Model 3 while *F*-test statistics for *MPAA* ratings are 1.91 for *ex ante* Model 1, 1.59 for *ex post* Model 2 and 1.12 for *ex post* Model 3). De Vany (2004) confirms the significance of dummy variables for genres and *MPAA* ratings.

Model 1 – *ex ante* model.

$$\begin{aligned}\log(GROSS) &= \beta_0+\beta_1[\log(BUDGET)]+\beta_2(CRITIC)\\&+\beta_3[\log(ORIGGROSS)]+\beta_4[\log(INCOME)]\\&+\beta_5(CPITICKET)+\beta_6(PREVAUD)\\&+\beta_7(SUMMER)+\beta_8(THANKS/XMAS)\\&+\beta_9(STARPOWR2)+\beta_{10}(ACTADV)\\&+\beta_{11}(ANIM)+\beta_{12}(COMDRA)+\beta_{13}(COMEDY)\\&+\beta_{14}(HORROR)+\beta_{15}(SCIFI)+\beta_{16}(G)+\beta_{17}(PG)\\&+\beta_{18}(PG13)+\beta_{19}(DUM98)+\beta_{20}(DUM99)\\&+\beta_{21}(DUM00)+\beta_{20}(DUM01)+e \qquad (1)\end{aligned}$$

Model 2 – *ex post* model.

$$
\begin{aligned}
&\log(GROSS)\\
&\quad = \beta_0 + \beta_1[\log(BUDGET)] + \beta_2(CRITIC)\\
&\quad + \beta_3[\log(ORIGGROSS)] + \beta_4[\log(INCOME)]\\
&\quad + \beta_5(CPITICKET) + \beta_6(PREVAUD)\\
&\quad + \beta_7(SUMMER) + \beta_8(THANKS/XMAS)\\
&\quad + \beta_9(STARPOWR2) + \beta_{10}(CINEMASCORE)\\
&\quad + \beta_{11}(FILMNOM) + \beta_{12}(ACTADV) + \beta_{13}(ANIM)\\
&\quad + \beta_{14}(COMDRA) + \beta_{15}(COMEDY)\\
&\quad + \beta_{16}(HORROR) + \beta_{17}(SCIFI) + \beta_{18}(G) + \beta_{19}(PG)\\
&\quad + \beta_{20}(PG13) + \beta_{21}(DUM98) + \beta_{22}(DUM99)\\
&\quad + \beta_{23}(DUM00) + \beta_{24}(DUM01) + e \qquad (2)
\end{aligned}
$$

Model 3 – *ex post* model.

$$
\begin{aligned}
&\log(GROSS)\\
&\quad = \beta_0 + \beta_1[\log(SCREENS)] + \beta_2(CRITIC)\\
&\quad + \beta_3[\log(ORIGGROSS)] + \beta_4[\log(INCOME)]\\
&\quad + \beta_5(CPITICKET) + \beta_6(PREVAUD)\\
&\quad + \beta_7(SUMMER) + \beta_8(THANKS/XMAS)\\
&\quad + \beta_9(STARPOWR2) + \beta_{10}(CINEMASCORE)\\
&\quad + \beta_{11}(FILMNOM) + \beta_{12}(ACTADV) + \beta_{13}(ANIM)\\
&\quad + \beta_{14}(COMDRA) + \beta_{15}(COMEDY)\\
&\quad + \beta_{16}(HORROR) + \beta_{17}(SCIFI) + \beta_{18}(G) + \beta_{19}(PG)\\
&\quad + \beta_{20}(PG13) + \beta_{21}(DUM98) + \beta_{22}(DUM99)\\
&\quad + \beta_{23}(DUM00) + \beta_{24}(DUM01) + e \qquad (3)
\end{aligned}
$$

Table 3. *Ex ante* model 1 (available prior to release)

Dependent variable: log(*GROSS*)
$N = 438$

Variable	Coefficient	SE	*T*-stat
log(*BUDGET*)	0.3748***	0.0419	8.953
log(*ORIGGROSS*)	0.0162***	0.0063	2.598
log(*INCOME*)	4.5684*	2.7694	1.650
CPITICKET	−0.0147	0.0098	−1.495
CRITIC	0.0038***	0.0005	7.733
PREVAUD	−0.0027	0.0263	−0.104
SUMMER	0.1440***	0.0310	4.653
THANKS/XMAS	0.0696**	0.0311	2.240
STARPOWR2	0.0160	0.0099	1.619
ACTADV	0.1196***	0.0378	3.164
ANIM	0.0035	0.0708	0.050
COMDRA	0.0824**	0.0413	1.998
COMEDY	0.1072***	0.0356	3.008
HORROR	0.1316**	0.0538	2.446
SCIFI	0.0354	0.0646	0.548
G	0.0824	0.0792	1.040
PG	0.0361	0.0427	0.845
PG13	0.0638**	0.0276	2.316
DUM98	−0.1650**	0.0786	−2.099
DUM99	−0.0983	0.1089	−0.903
DUM00	−0.1232	0.1717	−0.718
DUM01	−0.0918	0.1984	−0.463
Constant	−52.706	34.931	−1.509

Notes: *, ** and *** indicate the significance at 10, 5 and 1% levels, respectively.
Adjusted $R^2 = 0.410$.

V. Results

Model 1 – ex ante results

Table 3 presents the results for the *ex ante* Model 1, which analyses 438 observations. The adjusted R^2 for this regression is 0.41. As hypothesized, log(*BUDGET*) is found to be positive and significant. All else equal, a 1% increase in a film's budget will correspond with an increase of 0.375% in revenue. De Vany (2004) also obtained point estimate elasticities of $<1\%$ (approximately 0.55). Such results indicate decreasing returns to a studio's film investment, and demonstrate revenue's inelasticity relative to changes in the budget variable. De Vany (2004, p. 124) argues that the production budget has the ability to raise the minimum revenue that a movie might earn, but has relatively small influence on its potential maximum. The budget achieves this by increasing the number of opening screens (i.e. the 'blockbuster strategy'), which increases the overall availability to consumers. The finding of decreasing returns belies the efficacy of the blockbuster strategy and supports the notion that budget alone is not enough to ensure a film's success. It is also likely that diminishing returns occur at a certain budget level, where the increase in spending is no longer obvious on the movie screen (De Vany, 2004, p. 134).

It is important to note that this study considers only the domestic market. Walls (2004) also found decreased domestic returns of \$0.98 in response to a \$1 increase in the production budget. His OLS regression does reveal, however, that a \$1 increase in the production budget increases the worldwide total by \$7.45. This demonstrates that one cannot reasonably assume that a loss in the domestic market translates to losses across the varying distribution systems. Foreign box-office receipts as well as the revenue obtained from DVD/VHS rentals and purchases and TV privileges undoubtedly raise the total profitability of motion pictures.

Both economic variables, log(*INCOME*) and *CPITICKET*, have the correct signs as hypothesized by economic theory. The positive and statistically significant coefficient on the income variable indicates that going to the movies is a normal leisure activity, *ceteris paribus*. Income elasticity is 4.6. The negative sign on the movie ticket variable, though insignificant, supports the law of demand. All else being equal, the quantity demanded of movies decreases as its price increases. Both economic variables are retained in the final specification for these reasons.

Cameron (1990) and other studies note mixed results in obtaining statistically significant income elasticities. Our coefficient signs for income and ticket price are consistent with many other studies (Fernández and Baños, 1997; Bagella and Becchetti, 1999; MacMillan and Smith, 2001). Bagella and Becchetti (1999) also found statistically insignificant impacts of price, as experienced in this study. MacMillan and Smith (2001) revealed an insignificant relationship between income and cinema attendance, but obtain a significant negative relationship between price and cinema admissions. Fernández and Baños (1997) found statistically significant impacts of both ticket price and income on cinema attendance. The insignificance of the price variable in this model is most likely a reflection of inadequacies in the variable's measurement as opposed to a lack of theoretical pertinence.

The coefficient on the *CRITIC* variable is positive and significant, which is consistent with the expectations and results of Sawhney and Eliashberg (1996) and Prag and Casavant (1994). The *STARPOWR*2 coefficient is positive but not significant. This outcome is supportive of the notion that stars themselves do not actually cause movies to be successful, but stars select film projects that already have an increased potential for success. De Vany (2005, p. 5) stated that '(m)ovies make stars, not the other way around'. Alternatively, stars may assist in 'rolling out' a film in the same fashion as De Vany (2004) claims that budget can be a component of studios' blockbuster strategies. Both De Vany (2004) and Prag and Casavant (1994) obtained positive and significant coefficients for their linear measures of stardom. As in this study, Ravid (2004) also found an insignificant result for stardom.

The estimated coefficients are positive and significant for both seasonal release variables, as anticipated. Films released during the summer and holiday seasons experience 14.4 and 7.0% higher respective revenues than other films, *ceteris paribus*. Simonoff and Sparrow (2000) obtains comparable results with a similar variable. The 1998 dummy is the only statistically significant year. It is concluded that films released in 1998, on average, were likely to do worse financially than films released in 1997. None of the other years has any unique factors that generate significantly higher or lower film revenue.

The log(*ORIGGROSS*) variable has a positive and significant coefficient. Based on these results, a 1% increase in the preceding film's gross will result in an increase of 0.016% in the sequel's gross revenue, all other things being equal. This is consistent with the findings of Prag and Casavant (1994) and De Vany (2004). This also supports the notion that sequels usually earn only a fraction of the gross of the original film.

Four of the six genre coefficients are positive and significant. The significance of *ACTADV* and *HORROR* are in accordance with the results of Ravid (2004). His results show that a consistent demand exists for violent films. Walls (2004, p. 94) also presented the notion that violent films, particularly mobster pictures, are able to transcend both cultural borders and time periods. *HORROR* has the largest coefficient of the genres, indicating a reliable audience unmatched by the others. Although inconsistent across their models, Bagella and Becchetti (1999) also found a positive impact of horror.

Comedies and comedy-dramas both have wide-reaching appeal, which leads to their significance. Bagella and Becchetti (1999) confirm the results regarding comedies for Italian audiences. Animated and science fiction features have a more hit-or-miss appeal, as demonstrated by their statistical insignificance. This may be attributed to the more limited target audiences of children's and science fiction films. Relatively rare films from this genre are able to transcend this barrier. For this reason, Ravid (2004) suggested that these films tend to be riskier undertakings for studios relative to violent, R-rated films.

The *PG*13 dummy is the only significant *MPAA* rating, whereas Prag and Casavant (1994) found positive and significant results for all ratings. The target audiences of *G* and *PG* films are younger, conservative, and make up a small proportion of the aggregate population. Like the children's genre audience, these groups are less consistent in their attendance. According to Prag and Casavant (1994), *PG*13 films tend to yield higher revenues due to a mature audience with higher levels of disposable income. Older moviegoers also attend evening viewings, which allows theaters more showings.

Table 4. *Ex post* model 2 (available after release)

Dependent variable: log(*GROSS*)
$N = 420$

Variable	Coefficient	SE	*T*-stat
log(*BUDGET*)	0.3609***	0.0431	8.360
log(*ORIGGROSS*)	0.0123**	0.0059	2.080
log(*INCOME*)	3.6008	2.6523	1.358
CPITICKET	−0.0105	0.0094	−1.112
CRITIC	0.0021***	0.0005	3.813
PREVAUD	0.0110	0.0252	0.439
SUMMER	0.1220***	0.0294	4.145
THANKS/XMAS	0.0503*	0.0305	1.649
STARPOWR2	0.0229**	0.0093	2.460
CINEMASCORE	0.0422***	0.0062	6.833
FILMNOM	0.0170***	0.0043	3.986
ACTADV	0.1293***	0.0356	3.632
ANIM	0.0290	0.0667	0.434
COMDRA	0.0341	0.0401	0.851
COMEDY	0.1297***	0.0344	3.772
HORROR	0.2030***	0.0515	3.939
SCIFI	0.1339**	0.0635	2.109
G	0.0107	0.0756	0.148
PG	−0.0128	0.0420	−0.304
PG13	0.0403	0.0270	1.494
*DUM98**	−0.1272	0.0756	−1.682
DUM99	−0.0637	0.1045	−0.610
DUM00	−0.1005	0.1643	−0.612
DUM01	−0.0799	0.1904	−0.420
Constant	−40.987	33.463	−1.225

Notes: *, ** and *** indicate the significance at 10, 5 and 1% levels, respectively.
Adjusted $R^2 = 0.486$.

Model 2 – ex post model results

Table 4 presents the results for *ex post* Model 2, which analyses a decreased number of 420 total observations.[11] The adjusted R^2 is 0.49. *DUM*98 remains negative and significant. Neither *INCOME* nor *CPITICKET* are statistically significant at conventional levels in this regression. None of the *MPAA* ratings are significant. The log(*BUDGET*), log(*ORIGGROSS*), *CRITIC* and the release date estimates remain positive and significant with coefficients of comparable magnitudes.

While the release dates do retain levels of significance, substantial differences are evident between the results of Models 1 and 2. *THANKS/XMAS* drops from a 5% level of significance to the 10% level. *SUMMER* remains at the 1% level but the coefficient of the former is smaller than that of the latter. One could argue that because the longer summer season is uninterrupted for 3 months, it has greater revenue earning potential than the shorter, interrupted thanksgiving and Christmas vacations. Holiday release films drop in revenue upon the termination of the holiday while a summer release film would not experience the seasonal conclusion drop until long after it had peaked and been replaced.

The *CRITIC* coefficient remains positive and significant, but with a smaller value of 0.0021. The results for the *CINEMASCORE* variable are positive and significant at the 1% level, as hypothesized. This shows that while positive critic opinions still encourage attendance in the post-release market, a consumer responds more to the word-of-mouth opinions of other consumers based on the size of the coefficient. For each incremental increase in the cinemascore grade, a 4% increase is expected in gross revenue, all else equal. It is therefore beneficial for films to impress audiences early in their run in order to gain strength and longevity.

The positive and statistically significant impact of *CINEMASCORE* is an important result. It establishes the crucial role of word-of-mouth information transfer. As predicted by De Vany (2004, p. 49), moviegoers 'can and do communicate their private evaluations about the quality of a movie'. When viewers share their personal opinions with others, De Vany's 'uninformative information' cascade breaks and becomes an informative information cascade. Following a movie's opening weekend, the influence of 'blockbuster strategy' variables such as budgets and stars, weakens substantially. The estimated coefficient of *CINEMASCORE* is much larger than *STARPOWR*2's and lends credence to this assertion.

There is a change in the significant genre variables. While *ACTADV*, *COMEDY* and *HORROR* retain positive and significant coefficients, *COMDRA* does not. Prag and Casavant (1994) find only one genre, drama, to be statistically significant and it is negative. The results for *HORROR* and *COMEDY* continue to reflect the findings of Bagella and Becchetti (1999). The *HORROR* variable benefits in the post-release market as well. Its larger coefficient continues to demonstrate likelihood that audiences in the horror genre build slowly and benefit from the flow of information exchange between consumers. The *SCIFI* variable is boosted in the *ex post* regression, developing significance at the 5% level.

The *t*-statistic of *STARPOWR*2 increases and becomes significant at the 5% level. This supports the findings of De Vany and Walls (1997) and Bagella and Becchetti (1999) and is predicted

[11] The reduced observations are a result of missing data for the *CINEMASCORE* variable.

by Rosen (1981). This may indicate that consumers who are persuaded by high profile stars are also slow to act on their desire to see new films. Sawhney and Eliashberg (1996) found a similar effect with R-rated films and films that contain high levels of sexual content. Albert (1998) confirmed the drawing power and marking power of stars in yielding successful films. De Vany (2004) suggested that stars might play a role in signalling film quality, but cautions that they only increase the probability of unlikely positive outcomes.

FILMNOM's estimate is positive and significant at the 1% level, with each additional film nomination leading to a 1.7% increase in gross revenue, *ceteris paribus*. Similarly, Smith and Smith (1986), Prag and Casavant (1994) and Deuchert *et al.* (2005) obtained a positive and significant impact of the total number of Academy Awards received by a film and its success.

Model 3 – ex post model results

Table 5 presents the results for *ex post* Model 3, where the adjusted R^2 is 0.65. This regression is based on more observations ($n = 466$) than either the *ex ante* Model 1 or *ex post* Model 2. The log(*ORIGGROSS*), *CRITIC*, and release date estimates remain positive and significant while *DUM*98 remains negative and significant. None of the *MPAA* ratings are significant. *SUMMER* remains significant at the 1% level while *THANKS*/*XMAS* increases in significance from the level of 10 to 1%. The results for the *CINEMASCORE* variable remain positive and significant at the 1% level with a coefficient of comparable magnitude. *STARPOWR*2 and *FILMNOM* remain positive and significant. *ACTADV* and *COMEDY* retain positive and significant coefficients. All other genres become statistically insignificant. The result for *COMEDY* continues to reflect the findings of Bagella and Becchetti (1999).

The estimated coefficient for log(*screens*) is positive and significant at the 1% level. The magnitude is almost four times greater than the coefficient of log(*budget*) in the previous *ex post* model. The elasticity of gross revenue with respect to peak screens is 1.5 whereas the elasticity of gross revenue with respect to production budget is 0.36. This suggests that film performance is very sensitive to the peak number of screens, on which it plays. The increasing returns to peak screens may support the notion that screens have the potential to 'prop up' the minimum revenue that a movie might earn. Simonoff and Sparrow (2000), MacMillan and Smith (2001) and De Vany (2004) obtained similar results in their studies. It is possible that some moviegoers attend films regardless of quality and view whatever is playing locally. This is similar to a sports fan who is likely to watch any game that is televised.

Table 5. *Ex post* model 3 (available after release)

Dependent variable: log(*GROSS*)
$N = 466$

Variable	Coefficient	SE	*T*-stat
log(*SCREENS*)	1.4976***	0.0858	17.462
log(*ORIGGROSS*)	0.0089*	0.0048	1.835
log(*INCOME*)	2.7454	1.9896	1.380
CPITICKET	−0.0028	0.0071	−0.398
CRITIC	0.0030***	0.0004	6.997
PREVAUD	0.0196	0.0206	0.955
SUMMER	0.1005***	0.0232	4.340
THANKS/*XMAS*	0.0752***	0.0249	3.018
STARPOWR2	0.0194**	0.0077	2.511
CINEMASCORE	0.0345***	0.0051	6.744
FILMNOM	0.0210***	0.0036	5.865
ACTADV	0.0731**	0.0290	2.520
ANIM	−0.0445	0.0557	−0.800
COMDRA	0.0056	0.0316	0.177
COMEDY	0.0595**	0.0264	2.254
HORROR	0.0493	0.0431	1.144
SCIFI	0.0789	0.0523	1.507
G	−0.0542	0.0636	−0.852
PG	−0.0226	0.0324	−0.699
PG13	0.0129	0.0218	0.594
DUM98	−0.1297**	0.0577	−2.248
DUM99	−0.1295	0.0822	−1.575
DUM00	−0.1819	0.1299	−1.400
DUM01	−0.2068	0.1511	−1.369
Constant	−39.964	25.109	−1.313

Notes: *, ** and *** indicate the significance at 10, 5 and 1% levels, respectively.
Adjusted $R^2 = 0.645$.

Further discussion of the models

It is apparent that Models 2 and 3 differ in two respects – the former includes log(*budget*) with an adjusted R^2 of 0.49 and the latter includes log(*screens*) instead with an adjusted R^2 of 0.65. Log(*budget*) and log(*screens*) are highly collinear, as suggested by De Vany's (2004) description of the 'uninformative information' strategy of studios. It is necessary to discern whether moviegoers are more attuned to a film's budget or its number of screens once it has been released. In order to determine the correct model specification, the Davidson–MacKinnon (1993) *J*-test for nonnested models is applied. The fitted dependent variable from Model 3 has significant explanatory power as an additional regressor in Model 2 (at the 5% level) while the fitted dependent variable from Model 2 lacks significant impact as an additional independent variable in Model 3. Therefore, Model 3 is determined to be the superior *ex post* specification.

The adjusted R^2 values obtained in this study lie at the middle to high ends of the ranges reported by other studies. For researchers reporting either unadjusted or adjusted R^2 measures for similar regression analyses, most values range from a low of approximately 0.09 to a high of about 0.7 with a general concentration of values in the vicinity of 0.3–0.4. Our results range between 0.4 and 0.65, which are consistent with the pre-existing literature and noteworthy considering the high variability in the dependent variable.

VI. Conclusion

Using information available prior to a film's release, the *ex ante* regression results indicate that production budget, the gross of a prequel, income, favourable critical reviews, summer releases and thanksgiving/Christmas releases exert positive and statistically significant effects on film success. The results of the *ex post* regressions demonstrate that peak screens, the quadratic of stardom, film award nominations and positive word-of-mouth information sharing are all positive and significant determinants of film performance. The impacts of production budget, the gross of a prequel, positive critical reviews, and release dates during the summer and thanksgiving/Christmas seasons are robust across the models. The action-adventure and comedy genres were also consistently positive and statistically significant across the models.

A few remarkable differences emerge when comparing the *ex ante* and *ex post* regressions. Based on the results, it appears that personal income, the comedy-drama genre, and the *PG*13 ratings are the important positive factors in the financial success of a film when moviegoers know relatively little about a picture. Once a film is released and more information is made available to customers, these variables cease to matter appreciably. The nonlinear effect of 'star power' becomes significant post-release. Perhaps most notably, the *ex post* results reveal the importance of award nomination and word-of-mouth praise in the financial success of films. It appears that indicators of film quality such as approbation from the motion picture industry and praise from peer moviegoers result in higher gross revenue.

Acknowledgements

We gratefully acknowledge the superb research assistance of Stephanie Perkovich. This study has benefited from suggestions offered by participants at the Pennsylvania Economic Association Conference.

References

Abbé-Decarroux, F. (1994) The perception of quality and the demand for services: empirical application to the performing arts, *Journal of Economic Behavior and Organization*, **23**, 99–107.

Albert, S. (1998) Movie stars and the distribution of financially successful films in the motion picture industry, *Journal of Cultural Economics*, **22**, 249–70.

Bagella, M. and Becchetti, L. (1999) The determinants of motion picture box office performance: evidence from movies produced in Italy, *Journal of Cultural Economics*, **23**, 237–56.

Becker, G. S. (1965) A theory of the allocation of time, *The Economic Journal*, **40**, 493–517.

Cameron, S. (1986) The supply and demand for cinema tickets: some U.K. evidence, *Journal of Cultural Economics*, **10**, 38–62.

Cameron, S. (1990) The demand for cinema in the United Kingdom, *Journal of Cultural Economics*, **14**, 35–47.

Davidson, R. and MacKinnon, J. G. (1993) *Estimation and Inference in Econometrics*, Oxford University Press, London.

Deuchert, E., Adjamah, K. and Pauly, F. (2005) For Oscar glory or Oscar money?, *Journal of Cultural Economics*, **29**, 159–76.

De Vany, A. (2004) *Hollywood Economics: How Extreme Uncertainty Shapes the Film Industry*, Routledge, New York.

De Vany, A. and Walls, D. (1996) Bose-Einstein dynamics and adaptive contracting in the motion picture industry, *The Economic Journal*, **439**, 1493–514.

De Vany, A. and Walls, D. (1997) The market for motion pictures: rank, revenue, and survival, *Economic Inquiry*, **35**, 783–97.

De Vany, A. and Walls, D. (1999) Uncertainty in the movie industry: does star power reduce the terror of the box-office?, *Journal of Cultural Economics*, **23**, 285–318.

Dewenter, R. and Westermann, M. (2005) Cinema demand in Germany, *Journal of Cultural Economics*, **29**, 213–31.

Eliashberg, J. and Shugan, S. M. (1997) Film critics: influencers or predictors?, *Journal of Marketing*, **61**, 68–78.

Fernández, B. V. and Baños, P. J. (1997) Cinema demand in Spain: a cointegration analysis, *Journal of Cultural Economics*, **21**, 57–75.

The Internet movie database [online], Internet Movie Database, Inc. Available from: http://pro.imdb.com (accessed 1 February 2003–28 May 2004).

Jansen, C. (2005) The performance of German motion pictures, profits and subsidies: some empirical evidence, *Journal of Cultural Economics*, **29**, 191–212.

Kelwick, J. (2002) The Usher – Just one opinion [online], Available from: http://www.kelwick.karoo.net/index.htm (accessed 15 September 2002–23 June 2004).

MacMillan, P. and Smith, I. (2001) Explaining post-war cinema attendance in Great Britain, *Journal of Cultural Economics*, **25**, 91–108.

Moore, T. G. (1966) The demand for Broadway theatre tickets, *Review of Economics and Statistics*, **48**, 79–87.

Nash, B. (2002) The numbers: Box office data, movie stars, idle speculation [online], Nash Information Services. Available from: http://www.the-numbers.com (accessed 26 September 2002–27 June 2004).

Prag, J. and Casavant, J. (1994) An empirical study of the determinants of revenues and marketing expenditures in the motion picture industry, *Journal of Cultural Economics*, **18**, 217–35.

Ravid, S. A. (1999) Information, blockbusters, and stars: a study of the film industry, *The Journal of Business*, **4**, 463–92.

Ravid, S. A. (2004) Are they all crazy or just risk averse? Some movie puzzles and possible solutions, in *Contributions to Economic Analysis*, Vol. 260 (Ed.) V. A. Ginsburgh, Elsevier, New York, NY, pp. 33–47.

Rosen, S. (1981) The economics of superstars, *The American Economic Review*, **71**, 845–58.

Sawhney, M. S. and Eliashberg, J. (1996) A parsimonious model for forecasting gross box-office revenues of motion pictures, *Marketing Science*, **15**, 113–31.

Simonoff, J. S. and Sparrow, I. R. (2000) Predicting movie grosses: winners and losers, blockbusters and sleepers, *Chance*, **13**, 15–24.

Smith, S. P. and Smith, V. K. (1986) Successful movies: a preliminary empirical analysis, *Applied Economics*, **18**, 501–7.

Vogel, H. L. (1990) *Entertainment Industry Economics: A Guide for Financial Analysis*, 2nd edn, Cambridge University Press, New York.

Walls, W. D. (2004) Revenues, profitability, and returns: clinical analysis of the market for mobster films, *International Journal of Business and Economics*, **2**, 93–106.

Withers, G. (1980) Unbalanced growth and the demand for performing arts: an econometric analysis, *Southern Economic Journal*, **46**, 735–42.

Appendix 1. Films included in sample

Titanic	Amistad	The Sixth Man
Men in Black	Soul Food	First Strike
The Lost World	Wag The Dog	In Love and War
Liar, Liar	The Devil's Own	Excess Baggage
Air Force One	Donnie Brasco	Saving Private Ryan
As Good As it Gets	The Peacemaker	Armageddon
Good Will Hunting	Private Parts	There's Something About Mary
My Best Friend's Wedding	Money Talks	A Bug's Life
Tomorrow Never Dies	Jackie Brown	The Waterboy
Face/Off	Seven Years in Tibet	Dr. Dolittle
Batman & Robin	National Lampoon's Vegas Vacation	Rush Hour
George of the Jungle	Mortal Kombat: Annihilation	Deep Impact
Scream 2	Selena	Godzilla
conAir	Addicted to Love	Patch Adams
Contact	The Relic	Lethal Weapon 4
Hercules	Metro	The Truman Show
Flubber	For Richer or Poorer	Mulan
Conspiracy Theory	Beverly Hills Ninja	You've Got Mail
I Know What you Did Last Summer	Picture Perfect	Enemy Of the State
Dante's Peak	Home Alone 3	The Prince of Egypt
Anaconda	Fools Rush In	The Rugrats Movie
L.A. Confidential	Romy and Michelle's High School Reunion	Shakespeare in Love
In & Out	Father's Day	The Mask of Zorro
The Fifth Element	Grosse Pointe Blank	Stepmom
Mouse Hunt	Out to Sea	Antz
The Saint	The Edge	The X-Files
The Devil's Advocate	Event Horizon	Wedding Singer
Kiss The Girls	An American Werewolf in Paris	City of Angels
Jungle To Jungle	Boogie Nights	The Horse Whisperer
Anastasia	Murder At 1600	Six Days, Seven Nights
Spawn	Mimic	Star Trek: Insurrection
The Jackal	Midnight in the Garden of Good and Evil	Blade
Starship Troopers	Air Bud	Lost In Space
Austin Powers: International Man of Mystery	Good Burger	A Perfect Murder

(*continued*)

Appendix 1. Continued

Breakdown	Hoodlum	The Parent Trap
Absolute Power	Red Corner	Ever After
The Game	Mr. Magoo	Hope Floats
G.I Jane	The Apostle	US Marshals
Speed 2: Cruise Control	Booty Call	Life is Beautiful
Alien Resurrection	Gone Fishin'	The Man In The Iron Mask
Volcano	That Darn Cat	A Civil Action
The Rainmaker	The Postman	Snake Eyes
The Full Monty	That Old Feeling	What Dreams May Come
Bean	Fire Down Below	Small Soldiers
Copland	Wes Craven Wishmaster	Halloween: H20
Nothing To Lose	RocketMan	Mighty Joe Young
Practical Magic	Dance With Me	Never Been Kissed
The Negotiator	Twilight	Forces of Nature
Meet Joe Black	Dead Man On Campus	Varsity Blues
Ronin	Soldier	Message in a Bottle
The Siege	Dark City	South Park: Bigger, Longer & Uncut
Pleasantville	Blues Brothers 2000	The Hurricane
The Faculty	Les Miserables	Stigmata
I Still Know What You Did Last Summer	Star Wars: The Phantom Menace	House on Haunted Hill
Primary Colors	The Sixth Sense	Anna and the King
Urban Legend	Toy Story 2	10 Things I Hate About You
How Stella Got Her Groove Back	Austin Powers: The Spy Who Shagged Me	Cruel Intentions
Out Of Sight	The Matrix	Fight Club
Sphere	Tarzan	My Favorite Martian
The Thin Red Line	Big Daddy	8MM
Jack Frost	The Mummy	For Love of the Game
Mercury Rising	Runaway Bride	Man on the Moon
Bride of Chucky	The Blair Witch Project	The Best Man
A Night at the Roxbury	Stuart Little	Instinct
Madeline	The Green Mile	Mickey Blue Eyes
Wild Things	American Beauty	The 13th Warrior
Spice World	The World Is Not Enough	October Sky
The Object of My Affection	Double Jeopardy	Lake Placid
The Big Hit	Notting Hill	Random Hearts
Paulie	Wild Wild West	Superstar
Bulworth	Analyse This	Dogma
Great Expectations	The General's Daughter	Mystery Men
Can't Hardly Wait	American Pie	The Insider
Fallen	Sleepy Hollow	Girl, Interrupted
The Avengers	Inspector Gadget	The Out-of-Towners
One True Thing	The Haunting	The Other Sister
Rounders	Entrapment	The Story of Us
Quest for Camelot	Pokemon: The First Movie	Baby Geniuses
Beloved	Payback	Blast from the Past
The Borrowers	The Talented Mr. Ripley	The Wood
He Got Game	Any Given Sunday	Arlington Road
Psycho	Deep Blue Sea	The Iron Giant
John Carpenter's Vampires	Galaxy Quest	Being John Malkovich
Mafia!	The Thomas Crown Affair	EDtv
Hard Rain	Blue Streak	At First Sight
Species II	End Of Days	Magnolia
The Replacement Killers	The Bone Collector	The Bachelor
The Odd Couple II	Bowfinger	Stir of Echoes
Simon Birch	Deuce Bigalow: Male Gigolo	Doug's 1st Movie
Babe: Pig in the City	Life	Summer of Sam
The Big Lebowski	She's All That	Anywhere But Here
Disturbing Behaviour	Three Kings	An Ideal Husband

(*continued*)

Appendix 1. Continued

Half Baked	Bicentennial Man	Drive Me Crazy
Rushmore	The Cider House Rules	The Rage: Carrie 2
54	Eyes Wide Shut	Go
Bringing Out the Dead	Mission to Mars	Billy Elliot
True Crime	Coyote Ugly	Battlefield Earth
Muppets From Space	Snow Day	Urban Legends: Final Cut
A Midsummer Night's Dream	The Whole Nine Yards	Boys and Girls
The Corruptor	Next Friday	Down to You
Election	Shanghai Noon	Wonder Boys
Music of the Heart	Romeo Must Die	Best in Show
Snow Falling on Cedars	Final Destination	The Ninth Gate
The Messenger: The Story of Joan of Arc	Finding Forrester	Harry Potter and the Sorcerer's Stone
Dr. Seuss' How the Grinch Stole Christmas	The Road to El Dorado	The Lord of the Rings: The Fellowship of the Ring
Cast Away	Men of Honor	Shrek
Mission: Impossible 2	Dude, Where's My Car?	Monsters, Inc.
Gladiator	The Tigger Movie	Rush Hour 2
What Women Want	O Brother, Where Art Thou?	The Mummy Returns
The Perfect Storm	Frequency	Pearl Harbor
Meet the Parents	The Replacements	Ocean's Eleven
X-Men	Pokemon: The Movie 2000	Jurassic Park III
Scary Movie	The Beach	Planet of the Apes
What Lies Beneath	Little Nicky	A Beautiful Mind
Dinosaur	Pitch Black	Hannibal
Crouching Tiger, Hidden Dragon	The Original Kings of Comedy	American Pie 2
Erin Brockovich	Bedazzled	The Fast and the Furious
Charlie's Angels	Autumn in New York	Lara Croft: Tomb Raider
Traffic	28 Days	Dr. Dolittle 2
Nutty Professor II: The Klumps	Keeping the Faith	Spy Kids
Big Momma's House	Bounce	Black Hawk Down
Remember the Titans	Hanging Up	The Princess Diaries
The Patriot	The Flintstones in Viva Rock Vegas	Vanilla Sky
Miss Congeniality	The Skulls	Legally Blonde
Chicken Run	Thirteen Days	The Others
Gone in Sixty Seconds	The 6th Day	America's Sweethearts
Unbreakable	My Dog Skip	Cats & Dogs
Me, Myself & Irene	Where the Heart Is	Save the Last Dance
Space Cowboys	Pay It Forward	Atlantis: The Lost Empire
The Emperor's New Groove	Wes Craven Presents: Dracula 2000	Jimmy Neutron: Boy Genius
Scream 3	Proof of Life	A.I. Artificial Intelligence
U-571	Return to Me	Training Day
Rugrats in Paris: The Movie	Almost Famous	Along Came a Spider
The Family Man	The Legend of Bagger Vance	Bridget Jones's Diary
Hollow Man	The Art of War	Scary Movie 2
Chocolat	Bless the Child	The Score
Shaft	The Watcher	Shallow Hal
Disney's The Kid	Love and Basketball	Swordfish
Road Trip	High Fidelity	The Mexican
Vertical Limit	Book of Shadows: Blair Witch 2	Down to Earth
Bring It On	The Adventures of Rocky & Bullwinkle	Spy Game
102 Dalmatians	Nurse Betty	The Wedding Planner
The Cell	Reindeer Games	Behind Enemy Lines
Rules of Engagement	Titan A.E.	Ali
Moulin Rouge	15 Minutes	
Rat Race	Hearts in Atlantis	
A Knight's Tale	Angel Eyes	
The Animal	Corky Romano	
Don't Say a Word	Heist	
Blow	Kingdom Come	
The Royal Tenenbaums	Joe Somebody	

(*continued*)

Appendix 1. Continued

Exit Wounds	Two Can Play That Game
Enemy at the Gates	Joy Ride
K-PAX	
Serendipity	
Kate & Leopold	
Domestic Disturbance	
Zoolander	
The One	
Thirteen Ghosts	
Bandits	
Gosford Park	
Heartbreakers	
I Am Sam	
Hardball	
Evolution	
Not Another Teen Movie	
Jeepers Creepers	
Kiss of the Dragon	
Recess: School's Out	
In the Bedroom	
Black Knight	
See Spot Run	
Amelie	
Driven	
What's the Worst That Could Happen?	
Final Fantasy: The Spirits Within	
From Hell	
Monster's Ball	
How High	
Snatch	
Jay and Silent Bob Strike Back	
Riding in Cars With Boys	
Double Take	
Baby Boy	
The Majestic	
The Brothers	
Someone Like You	
Joe Dirt	
The Musketeer	
Crocodile Dundee in Los Angeles	
Captain Corelli's Mandolin	
Memento	
Sweet November	

Appendix 2. Stars included in *STARPOWR* variable

Allen, Tim	Mac, Bernie
Aniston, Jennifer	Martin, Steve
Bergen, Candice	Messing, Debra
Berry, Halle	Moore, Demi
Brokaw, Tom	Newman, Paul
Bullock, Sandra	Nicholson, Jack
Cage, Nicholas	O'Donnell, Rosie
Carey, Drew	O'Reilly, Bill
Carson, Johnny	Phil, Dr.
Clooney, George	Philbin, Regis
Connery, Sean	Pitt, Brad
Cosby, Bill	Rather, Dan
Costner, Kevin	Redford, Robert
Couric, Katie	Robert DeNiro
Cruise, Tom	Roberts, Julia
DeNiro, Robert	Romano, Ray
DiCaprio, Leonardo	Ryan, Meg
Eastwood, Clint	Schwarzenegger, Arnold
Elfman, Jenna	Seagal, Steven
Flockhart, Calista	Seinfeld, Jerry
Ford, Harrison	Smith, Will
Foster, Jodie	Snipes, Wesley
Gere, Richard	Springer, Jerry
Gibson, Mel	Stallone, Sylvester
Goldberg, Whoopi	Stewart, Jimmy
Grammar, Kelsey	Streep, Meryl
Hanks, Tom	Washington, Denzel
Heaton, Patricia	Wayne, John
Hunt, Helen	Williams, Robin
LeBlanc, Matt	Willis, Bruce
Leno, Jay	Winfrey, Oprah
Letterman, David	Winslet, Kate
Lopez, Jennifer	
Louis-Dreyfus, Julia	